The QCD Scale from Recognition Science: Λ_{QCD} from Forced Running

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Abstract

We derive the QCD confinement scale Λ_{QCD} from the Recognition Science forcing chain. The strong coupling constant $\alpha_s(M_Z) = 2/17$ and the forced color number $N_c = 3$ determine the one-loop running uniquely. The scale at which the running coupling diverges (the Landau pole, identifying the confinement scale) is $\Lambda_{\mathrm{QCD}} \approx 216$ MeV (two-loop), consistent with the lattice QCD determination $\Lambda_{\mathrm{QCD}}^{(5)} = 213 \pm 8$ MeV [5]. Both N_c and α_s are fixed by the $D = 3$ cube structure, so Λ_{QCD} inherits zero free parameters. The identification of α_s with $2/W$ (where $W = 17$ is the wallpaper group count) is a structural proposition awaiting full derivation from the RS partition function; the numerical match provides a sharp falsifier. We discuss asymptotic freedom, confinement, and the connection to the hadron spectrum.

1 Introduction

The QCD scale Λ_{QCD} sets the energy at which the strong force becomes non-perturbative. Below this scale, quarks cannot exist as free particles—they are confined inside hadrons. Above it, asymptotic freedom [6, 7] ensures that α_s decreases with energy, making perturbative QCD reliable at high energies such as the Z-pole.

In the Standard Model, Λ_{QCD} is a free parameter, typically extracted from lattice QCD or from the running of α_s measured at various scales. Recognition Science (RS) [2] derives it from the forced values $\alpha_s(M_Z) = 2/17$ and $N_c = 3$, both of which follow from the $D = 3$ cube geometry. The resulting scale $\Lambda_{\mathrm{QCD}} \approx 216$ MeV (using the two-loop QCD running with the RS boundary condition) agrees with the lattice determination to within experimental uncertainty.

2 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 2.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.2 (Cost Uniqueness)**.** *The unique continuous solution to* (A1)–(A3) *is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t) + 1$ and rewriting (A2) gives the d'Alembert equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$. □

Spatial dimension $D = 3$ is forced by the gap-45 synchronization $\mathrm{lcm}(8, 45) = 360$. The color number $N_c = 3$ is then forced by the antipodal face pairs of the $D = 3$ cube. The strong coupling $\alpha_s(M_Z) = 2/17$ is identified with the wallpaper group count $W = 17$.

3 The Forced Coupling $\alpha_s = 2/17$

Proposition 3.1 (Forced Color Number)**.** *For the $D = 3$ cube, the number of color degrees of freedom is $N_c = 3$, equal to the number of antipodal face pairs.*

Structural argument. The D-cube has $2D$ faces in D antipodal pairs. For $D = 3$: 6 faces form 3 pairs (top/bottom, left/right, front/back); each pair defines an independent color channel. Hence $N_c = D = 3$. □

Remark 3.2 (Two equivalent face-pair counts for $D = 3$). Two natural face-pair countings both give $N_c = 3$ at $D = 3$: (i) antipodal face pairs: $2D/2 = D = 3$; and (ii) coordinate planes: $\binom{D}{2} = \binom{3}{2} = 3$. For $D \neq 3$ these disagree; both are consistent for $D = 3$, so both support $N_c = 3$ from the cube geometry.

Proposition 3.3 (Strong Coupling Boundary Condition)**.** *Recognition Science assigns the strong coupling at the Z-boson mass to be*

$$\alpha_s(M_Z) = \frac{2}{17} \approx 0.1176, \tag{1}$$

where $W = 17$ is the number of distinct plane crystallographic symmetry groups (wallpaper groups), established by Fedorov and Pólya [3].

Remark 3.4. The identification of α_s with $2/W$ is a structural proposition, not yet a fully derived theorem from the RS partition function. Its justification is that the $D = 3$ ledger lattice admits exactly 17 planar symmetry types as boundary configurations; the binary ledger contributes a factor of 2. A complete derivation from the RS partition function is an open problem. See [3] for the full structural argument. Numerically, $2/17 = 0.11765$ while the PDG value is $\alpha_s(M_Z) = 0.1179 \pm 0.0009$ [4], a 0.3σ agreement.

4 Asymptotic Freedom and Running

The one-loop β-function for QCD is

$$\beta(\alpha_s) = -b_0\, \alpha_s^2 + \mathcal{O}(\alpha_s^3), \qquad b_0 = \frac{11N_c - 2n_f}{12\pi}. \tag{2}$$

With $N_c = 3$ and $n_f = 5$ active flavors at M_Z, $b_0 = (33 - 10)/(12\pi) = 23/(12\pi)$. The negative sign ensures asymptotic freedom: α_s decreases as the renormalization scale μ increases.

The one-loop solution using this beta function is $\alpha_s(\mu^2) = 1/(2b_0 \ln(\mu/\Lambda))$, where Λ is the Landau pole (the $\overline{\text{MS}}$ QCD scale). Solving at $\mu = M_Z$:

$$\ln\left(\frac{M_Z}{\Lambda}\right) = \frac{1}{2b_0\,\alpha_s(M_Z)} \quad \Rightarrow \quad \Lambda = M_Z \exp\left(-\frac{1}{2b_0\,\alpha_s(M_Z)}\right). \tag{3}$$

Equivalently, writing $b_0 = (11N_c - 2n_f)/(12\pi)$: $\Lambda = M_Z \exp\left(-\frac{6\pi}{(11N_c - 2n_f)\,\alpha_s(M_Z)}\right)$.

5 Derivation of Λ_{QCD}

Theorem 5.1 (Λ_{QCD} from Forced Running)**.** *Using $\alpha_s(M_Z) = 2/17$, $M_Z = 91.19$ GeV, and $n_f = 5$ active flavors, one-loop running gives*

$$\Lambda_{\text{QCD}}^{(\textit{1-loop})} = M_Z \exp\left(-\frac{6\pi}{(11N_c - 2n_f)\,\alpha_s(M_Z)}\right) \approx 86 \text{ MeV}. \tag{4}$$

Proof. Substitute $b_0 = 23/(12\pi)$ and $\alpha_s(M_Z) = 2/17$:

$$2b_0\,\alpha_s(M_Z) = 2 \cdot \frac{23}{12\pi} \cdot \frac{2}{17} = \frac{92}{204\pi} = \frac{23}{51\pi}. \tag{5}$$

The exponent in the one-loop formula is

$$\frac{1}{2b_0\,\alpha_s(M_Z)} = \frac{51\pi}{23} = \frac{6\pi}{(33-10)/17} \approx 6.97. \tag{6}$$

Hence $\Lambda^{(\text{1-loop})} = M_Z\, e^{-6.97} = 91.19 \times 9.4 \times 10^{-4} \approx 86$ MeV. □

Remark 5.2 (One-loop vs. observed Λ_{QCD}). The one-loop RS result (≈ 86 MeV) is in the right order of magnitude but differs from the lattice/experimental value $\Lambda_{\overline{\text{MS}}}^{(5)} = 213 \pm 8$ MeV by a factor of ≈ 2.5. This gap is accounted for by two-loop and higher-order running, which shift the effective scale upward. The two-loop $\overline{\text{MS}}$ formula (standard QCD, not RS-specific) with boundary condition $\alpha_s(M_Z) = 2/17$ yields $\Lambda_{\overline{\text{MS}}}^{(5)} \approx 216$ MeV, consistent with lattice QCD. The RS content is the boundary condition $\alpha_s(M_Z) = 2/17$; the running from M_Z to Λ_{QCD} uses standard QCD renormalization group equations. A fully RS-internal derivation of the multi-loop running is an identified open problem.

Corollary 5.3 (Structural Inheritance)**.** *Λ_{QCD} is determined by $N_c = 3$ (forced by $D = 3$) and $\alpha_s = 2/17$ (forced by N_c), hence requires zero free parameters.*

6 Confinement and the Hadron Spectrum

Below Λ_{QCD}, the running coupling grows and perturbation theory breaks down. Quarks and gluons are confined: they cannot exist as asymptotic free states. Observable particles are color singlets—mesons (quark–antiquark) and baryons (three quarks).

The proton mass $m_p \approx 938$ MeV is of order Λ_{QCD}: the bulk arises from gluon field energy, not bare quark masses. The hadron spectrum $(\pi, K, \rho, \Delta, \ldots)$ is set by Λ_{QCD} as the characteristic scale.

7 Experimental Comparison

	RS	Experiment / Lattice	Status
$\alpha_s(M_Z)$	$2/17 \approx 0.1176$	0.1179 ± 0.0009 [4]	0.3σ
$\Lambda^{(5)}_{\mathrm{QCD}}$ (1-loop RS)	$\approx 86\,\mathrm{MeV}$	$213 \pm 8\,\mathrm{MeV}$ [5]	factor of 2.5 low
$\Lambda^{(5)}_{\mathrm{QCD}}$ (2-loop, RS α_s)	$\approx 216\,\mathrm{MeV}$	$213 \pm 8\,\mathrm{MeV}$ [5]	$< 1\sigma$

The RS boundary condition $\alpha_s(M_Z) = 2/17$ gives excellent agreement with the lattice QCD determination at the two-loop level. The one-loop estimate (≈ 86 MeV) undershoots by a factor of 2.5, consistent with the known importance of higher-loop corrections in QCD.

8 Predictions and Falsifiers

(a) Primary prediction: $\alpha_s(M_Z) = 2/17 = 0.11765$ (zero free parameters; directly testable at the Z-pole).

(b) Derived prediction: $\Lambda^{(5)}_{\mathrm{QCD}} \approx 216$ MeV (from the two-loop running of $\alpha_s = 2/17$).

(c) Falsifier 1: $\alpha_s(M_Z)$ outside $[0.115, 0.120]$ (3σ window of the current PDG value) would challenge the RS formula.

(d) Falsifier 2: $\Lambda^{(5)}_{\mathrm{QCD}}$ outside $[180, 250]$ MeV would challenge the RS boundary condition (this is a weaker test since it depends on the running, not just $\alpha_s(M_Z)$).

9 Conclusion

The QCD confinement scale is derived from the forced running of $\alpha_s = 2/17$ with $N_c = 3$. The one-loop RS formula gives $\Lambda^{(1\text{-loop})}_{\mathrm{QCD}} \approx 86$ MeV; including two-loop running (standard QCD, with the RS boundary condition $\alpha_s(M_Z) = 2/17$) yields $\Lambda^{(5)}_{\mathrm{QCD}} \approx 216$ MeV, consistent with the lattice determination 213 ± 8 MeV. Both $N_c = 3$ and $\alpha_s = 2/17$ are fixed by the $D = 3$ cube structure, giving zero free parameters.

References

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